

Math 242, S13
Quiz 1

Name: *Answer*
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given a first-order differential equation

$$y' + y \tan x = \cos x.$$

(a) Verify that $y(x) = (x + C) \cos x$ defines a one-parameter family of solutions of the given differential equation.

$$y(x) = (x + C) \cos x$$

$$y'(x) = \cos x + (x + C)(-\sin x) = \cos x - (x + C) \sin x$$

$$\begin{aligned} \Rightarrow y' + y \tan x &= \cos x - (x + C) \sin x + (x + C) \cos x \cdot \tan x \\ &= \cos x - (x + C) \sin x + (x + C) \sin x \\ &= \cos x \end{aligned}$$

Thus $y(x) = (x + C) \cos x$ defines a one-parameter family of solutions of differential equation $y' + y \tan x = \cos x$.

(b) Determine a value of the constant C so that $y(x)$ satisfies the initial condition $y(\pi) = 0$.

$$y(\pi) = 0$$

$$\Rightarrow (\pi + C) \cos \pi = 0$$

$$-(\pi + C) = 0$$

$$\pi + C = 0$$

$$\Rightarrow C = -\pi$$